

Software Requirements Specification

Jackfruit Phase-1 — Student Record System

Tanmay T A, PES1UG24CS492 Uttam B S, PES1UG24CS508
Vibha Venkatesh Vasisht, PES1UG24CS526

Generated 2026-09-13

Contents

1	Revision History	2
2	1. Introduction	3
2.1	1.1 Purpose	3
2.2	1.2 Intended Audience and Reading Suggestions	3
2.3	1.3 Product Scope	3
2.4	1.4 References	3
3	2. Overall Description	3
3.1	2.1 Product Perspective	3
3.2	2.2 Product Functions	3
3.3	2.3 User Classes and Characteristics	3
3.4	2.4 Operating Environment	3
3.5	2.5 Design and Implementation Constraints	3
3.6	2.6 Assumptions and Dependencies	3
4	3. External Interface Requirements	4
4.1	3.1 User Interfaces	4
4.2	3.2 Software Interfaces	4
4.3	3.3 Communications Interfaces	4
5	4. Analysis Models	4
6	5. System Features	4
6.1	5.1 Onboarding	4
6.1.1	5.1.1 Description and Priority	4
6.1.2	5.1.2 Stimulus/Response Sequences	4
6.1.3	5.1.3 Functional Requirements	4
6.2	5.2 Edit Records	4
6.2.1	5.2.1 Description and Priority	4
6.2.2	5.2.2 Stimulus/Response Sequences	4
6.2.3	5.2.3 Functional Requirements	4
6.3	5.3 Mark Attendance (with constraints)	5

6.3.1	5.3.1 Description and Priority	5
6.3.2	5.3.2 Stimulus/Response Sequences	5
6.3.3	5.3.3 Functional Requirements	5
6.4	5.4 Mark Grades (with rubric)	5
6.4.1	5.4.1 Description and Priority	5
6.4.2	5.4.2 Stimulus/Response Sequences	5
6.4.3	5.4.3 Functional Requirements	5
6.5	5.5 Retrieval	5
6.5.1	5.5.1 Description and Priority	5
6.5.2	5.5.2 Stimulus/Response Sequences	5
6.5.3	5.5.3 Functional Requirements	5
6.6	5.6 Role-Based Access	5
6.6.1	5.6.1 Description and Priority	5
6.6.2	5.6.2 Stimulus/Response Sequences	5
6.6.3	5.6.3 Functional Requirements	6
7	6. Other Nonfunctional Requirements	6
7.1	6.1 Performance Requirements	6
7.2	6.2 Safety Requirements	6
7.3	6.3 Security Requirements	6
7.4	6.4 Software Quality Attributes	6
7.5	6.5 Business Rules	6
8	7. Other Requirements	6
9	Appendix A: Glossary	6
10	Appendix B: Field Layouts	6
10.1	Student	6
11	Appendix C: Requirement Traceability Matrix	7
12	Appendix D: Stakeholder Discussion Log	7
12.1	Persona	7
12.2	Discussion 1 — Discovery Call (DATE)	7
12.3	Discussion 2 — Deployment & Constraints (DATE)	7
12.4	Discussion 3 — Sign-off (DATE)	7

1 Revision History

Name	Date	Reason For Changes	Version
		Initial draft	0.1

Add a row every time a substantive change is made (not every typo fix). Keep this current — it's a required section.

2 1. Introduction

2.1 1.1 Purpose

TODO

2.2 1.2 Intended Audience and Reading Suggestions

TODO

2.3 1.3 Product Scope

TODO

2.4 1.4 References

TODO

3 2. Overall Description

3.1 2.1 Product Perspective

TODO

3.2 2.2 Product Functions

- Onboarding (student/staff registration)
- Edit records
- Mark attendance (with constraints)
- Mark grades (with rubric)
- Retrieval / search
- Role-based access control

3.3 2.3 User Classes and Characteristics

TODO

3.4 2.4 Operating Environment

TODO

3.5 2.5 Design and Implementation Constraints

TODO

3.6 2.6 Assumptions and Dependencies

TODO

4 3. External Interface Requirements

4.1 3.1 User Interfaces

TODO

4.2 3.2 Software Interfaces

TODO

4.3 3.3 Communications Interfaces

TODO

5 4. Analysis Models

TODO

6 5. System Features

6.1 5.1 Onboarding

6.1.1 5.1.1 Description and Priority

TODO — priority: High/Medium/Low

6.1.2 5.1.2 Stimulus/Response Sequences

TODO

6.1.3 5.1.3 Functional Requirements

- REQ-ONB-1:
- REQ-ONB-2:

6.2 5.2 Edit Records

6.2.1 5.2.1 Description and Priority

TODO

6.2.2 5.2.2 Stimulus/Response Sequences

TODO

6.2.3 5.2.3 Functional Requirements

- REQ-REC-1:
- REQ-REC-2:

6.3 5.3 Mark Attendance (with constraints)

6.3.1 5.3.1 Description and Priority

TODO

6.3.2 5.3.2 Stimulus/Response Sequences

TODO

6.3.3 5.3.3 Functional Requirements

- REQ-ATT-1:
- REQ-ATT-2:

6.4 5.4 Mark Grades (with rubric)

6.4.1 5.4.1 Description and Priority

TODO

6.4.2 5.4.2 Stimulus/Response Sequences

TODO

6.4.3 5.4.3 Functional Requirements

- REQ-GRD-1:
- REQ-GRD-2:

6.5 5.5 Retrieval

6.5.1 5.5.1 Description and Priority

TODO

6.5.2 5.5.2 Stimulus/Response Sequences

TODO

6.5.3 5.5.3 Functional Requirements

- REQ-RET-1:
- REQ-RET-2:

6.6 5.6 Role-Based Access

6.6.1 5.6.1 Description and Priority

TODO

6.6.2 5.6.2 Stimulus/Response Sequences

TODO

6.6.3 5.6.3 Functional Requirements

- REQ-RBAC-1:
- REQ-RBAC-2:

7 6. Other Nonfunctional Requirements

7.1 6.1 Performance Requirements

TODO

7.2 6.2 Safety Requirements

TODO — likely N/A or brief for this project type, but don't just delete the section.

7.3 6.3 Security Requirements

TODO

7.4 6.4 Software Quality Attributes

TODO

7.5 6.5 Business Rules

TODO

8 7. Other Requirements

TODO

9 Appendix A: Glossary

Term	Definition
Student	
Section	
Term	
Rubric	

10 Appendix B: Field Layouts

10.1 Student

Field	Length	Data Type	Description	Is Mandatory

11 Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
--------	----------------	-------------------	------------------------	------------------	---------------------	--------------	---------------------

12 Appendix D: Stakeholder Discussion Log

12.1 Persona

Name / Role: TODO — e.g. “Dr. [Name], Vice-Principal (Academics)” **Context:** TODO — one line on the institution and why they need this system.

12.2 Discussion 1 — Discovery Call (DATE)

Topics covered: initial scope, user roles, must-have features.

Q: TODO **A:** TODO

Q: TODO **A:** TODO

Decisions made: - TODO

12.3 Discussion 2 — Deployment & Constraints (DATE)

Topics covered: self-hosted vs. cloud, budget, expected concurrent users, data sensitivity / retention.

Q: Should this be cloud-hosted or self-hosted? **A:** TODO — record the reasoning, not just the conclusion.

Q: What’s the expected scale (students, concurrent users)? **A:** TODO

Q: Any data privacy or retention constraints? **A:** TODO

Decisions made: - Deployment model: TODO - Tech stack: TODO

12.4 Discussion 3 — Sign-off (DATE)

Topics covered: final scope confirmation, any descope features.

Q: TODO **A:** TODO

Decisions made: - TODO